Jyotil Agrawal

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Summary

Computer Engineering student with experience in software engineering, machine learning, embedded systems, and product development. Builds practical systems across data, software, and hardware using Python, C/C++, JavaScript/TypeScript, APIs, and cloud tools.

Education

Georgia Institute of Technology Atlanta, GA
B.S. Computer Engineering *Expected May 2028*
Relevant Coursework: Data Structures & Algorithms, Software Engineering, Statistics, Linear Algebra, Embedded Systems, Computer Architecture, Digital Design Laboratory

Experience

Undergraduate Researcher Sep 2026 - Present
Robotics & Intelligent Construction Automation Lab (RICAL), Georgia Tech *Atlanta, GA*
- Train and evaluate machine-learning models for AwareSite to recognize complex motions from cycles of simpler movements.
- Build Python and C pipelines for sensor-data collection, preprocessing, model training, validation, and performance analysis.
- Contribute to a separate research project under NDA focused on vehicle detection and highway safety.
- Automated data processing for 20 participants, reducing processing from 4+ days to hours and collection time from 60 to 30 minutes.

Full Stack and Product Intern Jun 2024 - May 2026
PrioriWise *Remote*
- Developed and released 25+ features using JavaScript/TypeScript, Node.js, MongoDB, REST APIs, and Git.
- Resolved 100+ issues through debugging, unit and integration testing, manual QA, and release validation; reduced deployment delays by 30%+.
- Led Agile sprints for a five-person team and built a human-reviewed AI workflow that reduced content production from 60 to 10 minutes per post.

Robotics Systems Intern Jan 2024 - May 2024
System Technology Works *Peachtree Corners, GA*
- Programmed robot functions in Python on NVIDIA Jetson Nano and embedded Linux, integrating movement, speech, sensors, motors, and AI.
- Analyzed 100+ validation datasets using pandas, NumPy, Excel, and serial debugging to identify hardware and software behavior issues.
- Reduced motor and behavior sequencing errors by about 30% and authored testing, setup, and troubleshooting documentation.

Projects

Secure NFC PCB Prototype | KiCad, NXP SE051A2, Raspberry Pi, I2C, Embedded Linux 2026 - Present
- Designed a fabrication-ready two-layer PCB with a secure element and 13.56 MHz NFC antenna.
- Validated Raspberry Pi communication over I2C and received a valid ATR at address 0x48 using Plug & Trust.

Sun-Tracking Solar Panel System | C/C++, Python, Sensors, Control Systems 2024 - 2025
- Built and tested a sensor-driven tracking system that increased measured energy generation by 57% versus a fixed-panel baseline across three trials.

Leadership

Venture Mentor 3 Years
Harvard Extension School Summer Entrepreneurship Program *Cambridge, MA*
- Selected as a mentor based on renewable-energy experience and advised three early-stage venture teams.
- Guided a solar venture through customer discovery, MVP scoping, testing, and product strategy; the team later secured a $2M test-site and funding contract.

Technical Skills

Languages: Python, C, C++, Java, SQL, JavaScript, TypeScript, Bash, R
Software & Data: Node.js, REST APIs, React, pandas, NumPy, scikit-learn, PyTorch, MongoDB, PostgreSQL, Jupyter
Embedded & Cloud: Embedded Linux, NVIDIA Jetson Nano, Raspberry Pi, Sensors, I2C, Serial Communication, AWS
Tools & Practices: Git/GitHub, CI/CD, Jira, Agile/Scrum, Unit/Integration Testing, Technical Documentation

Certification

Google Project Management Professional Certificate - Google Feb 2026